Case Study 2: Optimizing Course Scheduling at the Claremont Colleges Using Integer Programming

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1 Abstract

Scheduling can be a source of stress for Harvey Mudd students. Between balancing requirements and workload, multiple course sections, and overlapping courses, the number of different possible course schedules can be overwhelming. Through this project, we aim to create models that can help students find schedule choices that match their needs and reduce the time that students need to spend manually creating schedule options. Each model will generate a recommended list of courses (with section numbers) for a student of the 5C's in the Fall 2025 semester, based on their requirements and preferences. In this report we analyze two principal model formulations which reflect different popular concerns: Distance Minimization, with parameters that focus on accessibility, and Schedule Shape Optimization. Both of these formulations extend a base model which we title Student Interest Maximization, and we use them to meet students' common logistical and accessibility concerns. In order to combine the objective function of the base model with the extension, we establish a parameter to weight the consideration of the extension in comparison to the base. We investigate how changes to the weight parameter in these combined objective functions affects the resulting schedule in each case. Through this analysis, we find that the base model can create a balanced schedule, fulfilling major requirements and HSA interests with a moderate weight input on major courses. With each additive parameter in successive models, a penalty of around the same weight as the major course weight per credit hour is recommended to maintain a balance between overall student interest, major requirements, schedule timing, and building accessibility.

2 Executive Summary

Through this project, we created several different models to generate schedules that balance student preferences when it comes to course scheduling. Each model outputs a recommended list of courses (with the course number and section) for a student to enroll in for the semester. As research and stakeholder engagement, we surveyed three of the many parties involved with course scheduling: students, professors, and ADA coordinators. We designed each of our models around what typical students look at when designing their course schedule, and made adjustments according to our stakeholder interviews. All of our models take in a list of strict "Core requirements" (either for their major or the Common Core) which must be taken in that semester, a list of "major requirements" (courses required for a student's major which do not necessarily have to be taken in this semester), and a list of HSA courses with rankings, representing how much the student wants to take a course. The major requirements and ranked HSA courses are not strictly enforced, but incentivized. Thus, to consider both of these lists, our model needs to balance these courses relative to each other. We want to make this balancing of priorities customizable to the students – for example, one student may need to take more HSAs to stay on track with their graduation requirements, whereas another

may want to take more courses for their major requirements. However, we also provide the student with a 'default' of a balancing value to equally prioritize HSAs and major requirements.

Additional factors that we considered to match student preferences were the distances between back-to-back courses and building accessibility, as well as the 'shape' of their schedule. We have created one model for each of these two factors that builds off of the standard interest model described above. Both of these models are independent of each other and do not take the other factor into consideration. We consider the distance of a given course from the next course if it starts soon after, as students often prefer to take courses that are within a reasonable walking distance during the passing period (eg. it can be hard to get to a Mudd course in the ten minute passing period when coming from a Pomona course). Not only do some students have preferences against having courses that would require a long walk in a short time, but students may have accessibility needs that make it more difficult for them to commute further distances repeatedly. For this reason, we created a model that allows a student to penalize the distance between time-adjacent courses. Students can adjust this penalty parameter to a lower value (or zero) if they don't mind walking further. Additionally, this model allows students to increase the amount of time that they need in between courses and to penalize buildings that are not accessible. The other model that we have created takes into account the 'shape' of the schedule that it creates. By this, we mean that students often have preferences for morning or evening schedules, especially if they have extracurriculars that block off certain times of the day. Additionally, from our student survey, we noticed some students prefer to have no courses on a certain day of the week: most notably, Friday. Lastly, one consideration from an accessibility standpoint is that some students may have medication that can only be effective during certain hours of the day. Thus, we created a model which can allow the student to determine the earliest and latest course times they are willing to have in their schedule, as well as to penalize having courses on a certain day if they wish to do so.

To evaluate these models, we created three example student profiles. These example students are: a HMC junior Math major student entering their fall semester, a HMC senior Engineering major entering their spring semester, and an undeclared HMC freshman entering their spring semester. We created a list of required courses that they must take that semester, and another list of possible courses to meet their major requirements. We also created a list of HSA courses that each student is interested in taking. We then were able to test our models and observe the sensitivity to the different parameter weights of different factors.

The conclusions that we were able to draw from the models we ran with this example student are the following:

- For the base student interest model, using a low weight for the 'major multiplier' (which scales the 'point' values that major courses contribute to the objective function with respect to HSAs) caused the model to prioritize HSA courses, and a higher weight successfully made the schedule composed of almost all major courses. An insight that we drew from this is the importance of this balancing parameter between HSAs and major requirements so that a student doesn't end up taking 6 math courses in one semester (unless they explicitly intend to). If we were to recommend a default parameter value to a student, we believe that the best value would be 3 ($m = 3$), in order to create a schedule that has a good balance of HSA and major courses.
- In the distance and accessibility model, we observed that even a low penalty ($\delta = 1$) was able to cause the schedule to make consecutive courses be near each other. Additionally, this model did allow for courses that are further apart (eg. different campuses) to be taken as long as the time between the courses was sufficient according to the 'time between parameter'.
- With the accessibility parameter being set on, the schedules generated by the models did not

include courses that took place in buildings that were not accessible.

- We noted that in the schedule shape optimization, a higher penalty for undesirable times (<5) was able to more able remove courses outside of the time block that the student gave from the schedule as compared to a lower penalty (eg., 1), which weakly reduced the number of courses that violated the morning and night times provided.
- Additionally, we observed that when inputs to the model are more strict with respect to which courses need to be taken (for example, in the freshman spring semester when the Core courses are already determined for the student), there is less variation in the outputted schedules by each of the models even when varying the parameter values because there are less possible schedules overall.

3 Technical Report

3.1 Introduction

Each semester, students face the complex task of designing a course schedule to balance academic requirements, personal interests, and logistics. With hundreds of courses offered across the 5Cs, each with multiple sections, meeting times, locations, and professors, planning a course schedule can be overwhelming.

To address these conflicts, we created a base model that takes as input a student's list of HSA courses along with a ranking of their personal interest in each course. The student also inputs a list of major courses they would be interested in taking, and any courses they must take that semester. The model considers the practical constraints students face when designing a schedule, such as overlapping course times and credit limits, and outputs a feasible schedule that maximizes the student's overall cumulative interest in the scheduled courses. We represent the preference for a course through weighted values, the practical limitations through linear constraints, and formulate the problem as a mixed-integer linear program, which allows us to find an optimal schedule that satisfies all requirements. Building on the base model, we developed 3 additional formulations to reflect different personal priorities: minimizing the walking distance between courses, courses with ADA accessible buildings, and course meeting time preferences.

A local tool that many 5C students turn to in the creation of their schedules is Hyperschedule, a self-described "student-run course scheduler for the Claremont Colleges" [1]. We source our Spring and Fall 2025 course data from Hyperschedule's GitHub page. The data includes fields such as course codes, course area, meeting times, instructor name, campus affiliation, building location, and credit hours. We then formatted this data to import into our AMPL formulation

In order to run the models, we use three unique student profiles, each with different majors, class years, and preferences (see Appendix 4.6). The remainder of the report examines each of our model extensions, describing their functions and simplifying assumptions. The specific algebraic formulation for each is given in the Appendices. Next, we describe the results of a sensitivity analysis through different parameter values of our models. The recommended schedules produced are displayed as how students would view it on Hyperschedule. Finally, we provide our conclusions based on our holistic evaluation of the outputted schedules, as representatives of the stakeholder group. That is, we observe which models, using what values of weighting parameters, result in the most desirable schedules in our opinion. From this, we make a recommendation for the best-performing model and accompanying weighting parameters.

3.2 Stakeholder Engagement

Though Claremont College students are our main stakeholders in consideration, to gain an understanding of the overarching system that students interact with while scheduling their courses, we employed a three-pronged approach to our stakeholder engagement research. First, we surveyed the student body (see Appendix 4.1) to learn what considerations were most and least important to them when creating their course schedules. Second, we interviewed professors (see Appendix 4.2) to learn what they take into consideration while advising students through course scheduling. Lastly, we contacted the accessibility coordinators of the Claremont Colleges to gather information about the accessibility of buildings across the 5Cs, and learn more about the course scheduling process for students with various accommodations.

3.2.1 Student Survey Results

We received 64 responses in total to the student survey, which was sent to four Harvey Mudd dorm emailing lists: Linde Dorm, Case Dorm, Drinkward Dorm, and Atwood Dorm. From the numeric questions, we derived two main conclusions which informed changes to our model. Firstly, the two questions with the highest variance (out of the 1-5 rating questions) were our questions asking students whether they prefer to avoid morning or evening courses (2 and 3), which had variances of ≈ 1.52 and 1.57, respectively (App. 4.1). At first, we had intended to design our "Schedule Shape" model to rigidly penalize only morning or evening courses. However, given this near-equal spread of responses for each, we decided instead to create parameters for the start and end time of a penalized window, to allow students full freedom over the period they want to avoid. Secondly, we observed from the responses to questions 6 and 7 about distance and accessibility that the majority of our responders did not place much importance on distance, and even fewer took accessibility into account at all, with average responses of 2.05 and 1.69 respectively. However, the variances for these questions were not the lowest of the group, at ≈ 1.28 and 1.01 respectively (App. 4.1). So, we had motivation to implement our distance and accessibility models with editable weighting parameters, rather than predetermining the weighting for either.

3.2.2 Communication with Professors

We were able to get the input of four professors, which means that we cannot make any broad generalizations about these responses. However, what was echoed in these responses were three considerations commonly heard by advisors from students: major and Core requirements, overlapping meeting times, and personal interest. We were heartened to see these results, as we address all of these through different aspects of our models. Thus, we found no obvious edits to make to our models based on these insights.

3.2.3 Communication with Accessibility Coordinators

In order to gather information from all of the accessibility coordinators, we first sent an inquiry over email for any details about building accessibility that they had on record. Each of the coordinators responded with the interactive maps for their respective campus, which included a filter to display all ADA accessible entrances, elevators, and in some cases, ramps. From this data, we were able to create a parameter for each building, indicating whether the building has an accessible entrance.

To allow us to learn more about the experiences specific to Harvey Mudd students with accessibility needs, we conducted an individual interview with Harvey Mudd's ADA coordinator. During this interview, we were informed of several procedural pain points which informed our accessibility-focused model design. The first and main pain point within course scheduling for students with accessibility needs is the passing period. On Harvey Mudd's campus, for several of the academic

buildings, there is one ADA accessible entrance to the Parsons engineering building, which leads to one ADA accessible elevator. Then, for students to get to any of Galileo, McGregor, Jacobs, or Keck, students must navigate through the underground pathways called the Libra Complex, and take an accessible elevator back up. For this reason, the 10 to 15 minute passing period (which is already cumbersome for cross-campus commutes) can be entirely infeasible for students with accessibility needs. With this insight, we added a function to our model which would strongly penalize distance specifically between back-to-back courses, to give students plenty of time to travel.

We also gained insight into the course scheduling experience for those with ADA accommodations overall. Harvey Mudd students often come up with their schedules on their own time, and must then self-advocate to identify any challenges they may have with the building, classroom, or overall schedule. Then, their ADA coordinator may communicate with the registrar or other campuses to ensure they receive accommodations. This pipeline gives us even more motivation to create a model which can generate schedules quickly, to empower students to be able to confirm accommodations early.

3.3 Model Formulations

3.3.1 Student Interest Maximization (Base Model)

The *Student Interest Maximization* model is our base model, which the other three models extend to factor in other preferences students may have. This model maximizes the students' interest by taking in user-assigned rankings for a subset of courses and choosing a combination with the highest combined interest. Courses that satisfy major requirements are prioritized by applying a scalar parameter to scale the number of 'points' they contribute to the objective function. All courses are scheduled to avoid conflicting course meeting times while ensuring the courseload falls within a credit minimum and maximum range.

In constructing our MIP, we made some simplifying assumptions to keep the formulation feasible and limited to the information we could scrape, while still reflecting reasonable student behaviour. First, we assumed that each course would account for 3 credits which follows the convention Harvey Mudd uses; the other colleges in the consortium use 1 credit conventions, but we chose to adopt Mudd's system to standardize the workload for all courses across departments. It is assumed that 0-credit courses are not accounted for, as they tend to be either Physical Education (PE) courses or labs. Because 0-credit labs are paired with separate courses, we assume that students do not want to sign up for labs for which they are not taking the related course, and so we eliminate them from consideration to avoid that fringe case. Because interest in a PE course is very much contingent on specific levels of physical ability, we find it unhelpful to allow the model to choose any of them. Finally, if left unrestricted, the model attempts to schedule as many 0-credit courses as possible, to boost the objective function without contributing to the credit cap. Thus, for all of these reasons, we exclude 0-credit courses.

Within our dataset, select courses are listed as running for zero time. That is, the start time is written as "00:00", and the end time as "00:00". This indicates that the course is either asynchronous, or will schedule meetings based on the availability of each student. Though unconventional, we chose to include courses marked "00:00 - 00:00" because the student will ultimately be free to fit the course into the rest of the generated schedule. Thus, it should not introduce any scheduling conflicts. Because we disregard 0-credit courses as stated above, these zero-time courses also should not create any loopholes within the model, as it will eventually be capped by the credit limit parameter. Therefore, we cannot get into any situations where the model schedules an absurd amount of zero-time courses.

Next, we don't require the model to register students for all their listed major courses. Rather, each included major course contributes positively to the objective function which scales with a scalar parameter we call majorMultiplier. Using this, we can edit how strongly the model prefers major courses over HSA or other courses. This introduces an opportunity for an extension which allows strictly required courses.

In order to indicate preference among selected HSA courses, the student inputs rankings of HSA courses from 1 to n, with n being the number of HSA options, as a parameter to the model. For the same course title with equivalent sections, an identical ranking is assigned, thus causing the alternate sections of a course to be worth the same amount of "interest" points. This holds even if another section, under the same name, but taught by a different professor, might contribute more professor-rating points. Rationally, we assumed that students would prefer to take only one section of each course, since multiple equivalent sections don't add any additional benefit, knowledge or requirement-wise.

The formulation of the model itself is outlined in Appendix 3.4.1, with description of the parameters and constraints.

3.3.2 Distance Minimization and Accessibility Model

The *Distance Minimization and Accessibility Schedule* model applies similar principles from our base model, such as avoiding scheduling conflicts and preventing credit overloads. However, this version also takes into account the walking distance to each course and seeks to minimize the commute time to and between buildings where courses are held. In order to create this model, we got the building that each course is held at from Hyperschedule and then collected latitude and longitude data for each building. Then, the pairwise Euclidean distance between each course's building is computed from the latitude and longitude coordinates. These distances are then incorporated into the model data and objective function, so that courses held farther from other courses carry a higher cost. Users may also enable an accessibility penalty for courses held in buildings without accessible entrances. This penalty only applies when the time between courses is shorter than 15 minutes, reflecting that students with mobility needs can use alternate routes or entrances, which may require more time. For example, students may enter Jacobs, which does not have an accessible entrance, through Galileo and pass through the Libra complex. This becomes less of a deciding factor if the time window is sufficient, but distance restrictions can still be captured through the distance penalty. This model formulation is outlined in Appendix 4.4, with a description of the parameters and constraints.

3.3.3 Schedule Shape Optimization

The *Schedule Shape Optimization* model also builds on the base framework, constrained by time conflicts and credit limits. It incorporates user preferences for course start and end times by applying a penalty to courses that fall outside a preferred time window. The objective function maximizes student course interests after applying penalties for time preferences. The model produces a course load that is most convenient for the student.

This type of optimization is especially important for students who take medication which loses effectiveness after a few hours, as they cannot take courses in the evening at all. It can also be important for students with immovable commitments, such as student-athletes, or those who work a job during the semester. These types of commitments are constant throughout the semester and students must build their schedule around them. This model formulation is outlined in Appendix 4.5, with description of the parameters and constraints.

3.4 Results

For all runs of each of these models, credits are calculated according to the Harvey Mudd system, with minimum and maximum credit requirements of 12 and 18 credits. Looking at the generated schedule, we can see that it does prioritize the major requirements as we expected with the major multiplier, m = 5. The major multiplier is used as a balancer to prioritize major requirements relative to HSA courses.

3.4.1 Base Model Sensitivity Analysis

From this input, student interest maximization (base model) **3.3.1** outputs the following schedule, which can be depicted on the Hyperschedule interface (Fig. 1).

Similarly, using the student interest base model formulation, 2 alternative HMC majors' schedules for Fall 2025 were generated (Fig. 4). However, this time, the major multiplier was set to 3 and 1 respectively to see the impact on the generated schedules of figures 2, 3, 5, and 6.

Overall, there is no significant change in schedules except for the order of major courses when the major multiplier is changed. This demonstrates how altering the weight on major courses can influence course selection, while still respecting credit bounds and scheduling constraints.

Adjusting the major multiplier to m = 3 for the same major course requirements and HSA rankings generated a more balanced course load between HSA courses and major requirements (Fig. 2). However, the output from the *Student Interest Maximization* model with major multiplier m = 1, prioritized HSA courses over major requirements, producing a schedule heavily biased toward HSAs (Fig. 1). All outputs (Fig.1, 2, 3) incorporated the top choice HSA (MUS 096A), indicating that the most preferred HSA is scheduled consistently when its weight is equal to a major requirement.

This engineering student has been taking all their technical courses and needs their HSA breaths. Looking at the case where m = 3, we would get a model that balances their desire for HSA breaths as well as major requirements that they listed. If we upped the major requirement parameter to 5, the model expectly prioritizes courses listed in reqs over HSAs seen by the addition of ENGR151 and the loss of JAPN001B. On the other hand, if we set m = 1, the model lessens the prioritization of major requirements, and we see the addition of ECON125 and loss of ENGR133.

3.4.2 Distance Model Sensitivity Analysis

From this input, Student Interest Maximization **3.2.1** outputs the following schedules which can be represented on the Hyperschedule interface. For profiles 1 and 2, two separate inputs were considered: one focusing on minimizing overall distance, and another incorporating accessibility considerations at building entrances. For purely distance-focused inputs, high ($\delta = 5$) and low ($\delta = 1$) penalties were considered alongside a time window of 15 minutes. However, these yielded the same output schedules (Fig. 7, Fig. 9). In the accessibility-focused inputs, the distance penalty was increased to $\delta = 9$, and the allotted time window for cross-campus commutes was set to 30 minutes.

For our Math junior, both low ($\delta = 1$) and high ($\delta = 5$) penalties for distance and no accessibility factor ($y_c = 0$), creates a schedule such that students will have courses at multiple campuses within a day. For example, on Tuesdays and Thursdays, courses at both CMC and Pomona campuses are scheduled. Figure 7) represents the outputs for both $\delta = 1$ and 5, as the outputs were the same. However, we can see that the only directly consecutive courses, MATH144 and MATH171,

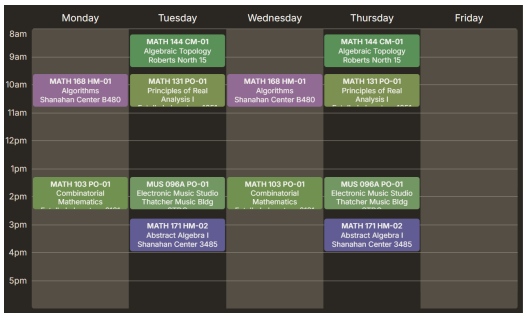


Figure 1: Profile 1 (Math Junior) Schedule generated by Student Interest Model 3.2.1 with m = 5.

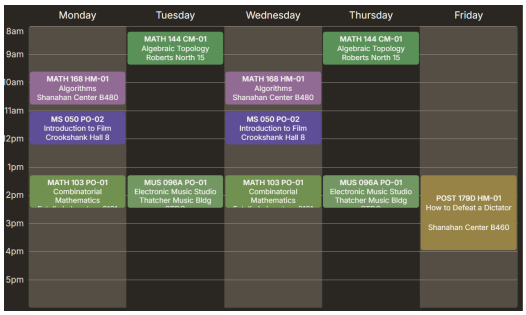


Figure 2: Profile 1 (Math Junior) Schedule generated by Student Interest Maximization Model 3.2.1 with m = 3.

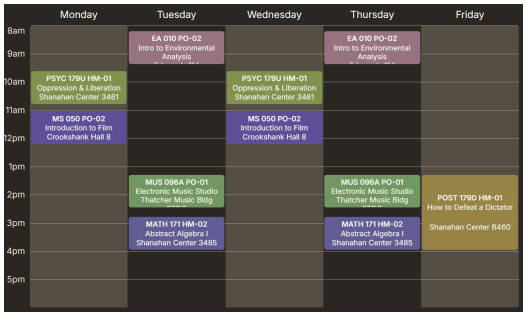


Figure 3: Profile 1 (Math Junior) Schedule generated by Student Interest Maximization Model 3.2.1 with m = 1.

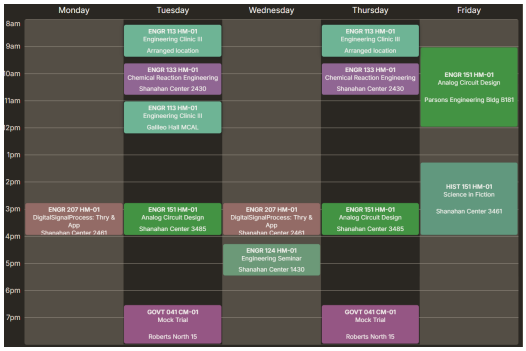


Figure 4: Profile 2 (Engineering Senior) Schedule generated by Student Interest Model 3.2.1 with m = 5.

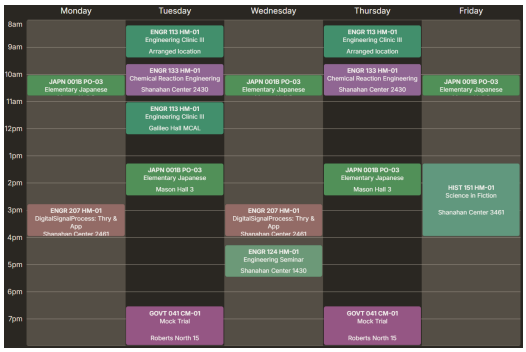


Figure 5: Profile 2 (Engineering Senior) Schedule generated by Student Interest Maximization Model 3.2.1 with m = 3.

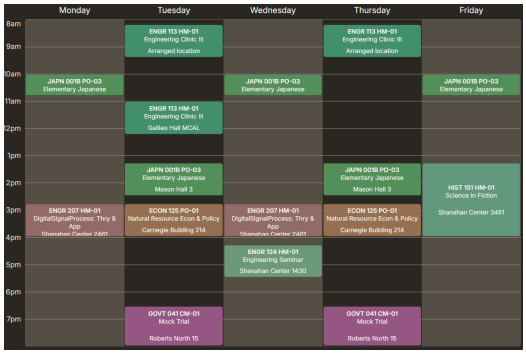


Figure 6: Profile 2 (Engineering Senior) Schedule generated by Student Interest Maximization Model 3.2.1 with m = 1.

are taught at the same campus, and the only cross-campus commute from CMC to Pomona has a window of over two hours. With the accessibility parameter active ($y_c = 3$) (Fig. 8), courses taught in an inaccessible building are excluded: MATH144 in Roberts North and MATH171 in Adams Hall, specifically. Both of these buildings lack an ADA accessible entrance, and are thus penalized. An

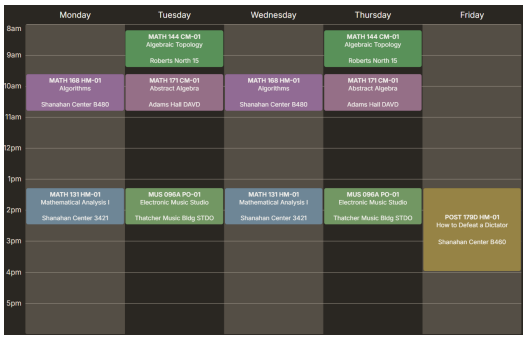


Figure 7: Profile 1 (Math Junior) Schedule generated by Distance Accessibility Model 3.2.1 with $m = 3, \delta = 1$ and $\delta = 5$, $y_c = 0$.

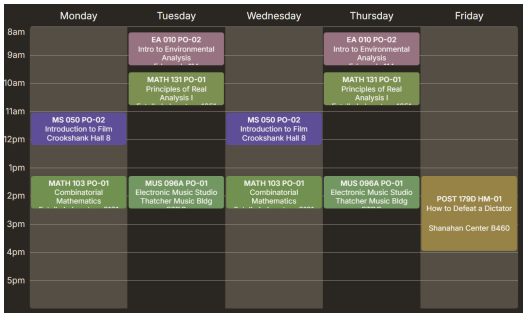


Figure 8: Profile 1 (Math Junior) Schedule generated by Distance Accessibility Model 3.2.1 with $m = 3, \delta = 9$, $y_c = 3$.

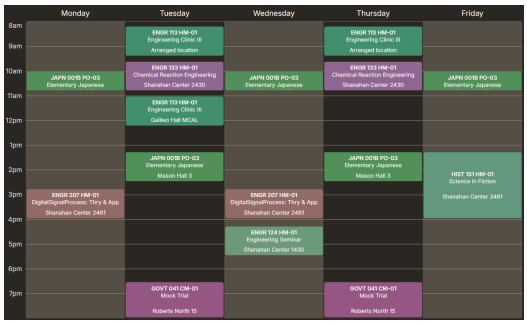


Figure 9: Profile 2 (Engineering Senior) Schedule generated by Distance Accessibility Model 3.2.1 with $m = 3, \delta = 1$, $\delta = 5$, and $\delta = 9$, $y_c = 0$ and $y_c = 3$.

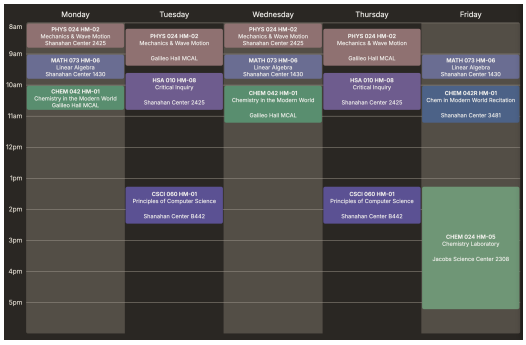


Figure 10: Profile 3 (Undecl. Freshman) Schedule generated by Distance Accessibility Model 3.2.1 with $m = 3, \delta = 1$, $\delta = 5$, and $\delta = 9$, $y_c = 0$ and $y_c = 3$.

interesting result from this optimization is that nearly the entire schedule in Fig. 8 is at Pomona. A quick review of Pomona's interactive accessibility map makes the reason obvious. Pomona has the highest density of identified ADA-accessible entrances out of the 5C's, and also hosts several Math courses which serve as suitable alternate for the Math junior. Thus, the model is able to select accessible classrooms, meet the major requirements, and minimize the walking distance all at once.

For our Engineering senior, the $\delta = 1$, 5, and 9 outputs were exactly the same. Thus, all three are shown in Fig. 9. With this configuration, with both $y_c = 0$ and $y = 3$, We see that, as required, there are no quick cross-campus commutes. We can analyze the course choices to see why outputs were identical. The generated schedule with the lowest of the parameters had already selected several accessible courses. The only inaccessible course is taught at Roberts North, and yet the course persists in all of the schedules. This course is rated as a 3 in the student's HSA rankings, which is exactly equal to y_c here. Thus, we can conclude that even an accessibility parameter value of $y_c = 3$ will give a generally accessible schedule, but may be overridden by the model to include HSA or major courses that may provide more utility to the model's output.

Finally, the Undeclared freshman profile showed no difference in output from the base model

under any parameter changes. Thus, only one figure is displayed (Fig. 10). This makes sense under all of our extensions, because a freshman at Harvey Mudd will naturally be taking nearly all of their classes at Harvey Mudd, minimizing the distance, and most of those classes are in Shanahan, which is ADA accessible. Thus, it makes sense that the optimal schedule for HSA interest and major requirement is identical to the optimal schedule for distance and accessibility.

3.4.3 Schedule Shape Sensitivity Analysis

We ran this model with all three of our profiles, at both a high and low weighting parameter constraining courses within the allowed time period, penalizing Friday courses according to the weighting parameter. Overall, we find that in these cases, the model did minimal shuffling of the schedules, and rather packed the schedule with low shape penalty (z = 1) full, and simply removed the non-required courses that did not fit the time period when a higher penalty was applied (z = 5).

For our Math Junior, we arbitrarily supposed that they are a morning person, and thus set the allowed window to 7am - 1pm. Fig. 11 shows the schedule output with z = 1. We can see that two courses past 1pm are scheduled, which is outside of the specified window. Notably, it also schedules three HSA courses from the students' HSA list, which is more than the average student would schedule in one semester. Fig. 14 shows the schedule output with z = 5, in which the two HSA's outside of the allowed window are dropped, and the one within the window is retained. This creates a compact schedule which allows the student to be done with their courses just after noon, with no Friday courses. The schedule of Fig. 11 did not change until z = 5 exactly—at z values from 1 to 4, the resulting schedules were the same.

The Engineering student was given a preferred evening window spanning from 2pm - 9pm. However, this run was much more constrained, as they have several required courses that fall outside of their preferred window. Fig. 12, representing z = 1, seems to fit everything it can without respect to the timing, similarly to the Math junior. This is consistent with the low weighting. In Fig. 15, we can see that the non-required morning courses are dropped, and a new, late afternoon HSA course is added to supplement. However, the Friday course is not avoided in this schedule, which reminds us that some schedule configurations will face tradeoffs even when optimized. Because GOVT041 only contributes 1.5 credits, and ENGR124 only 1 credit, that Friday course cannot be easily dropped, or else the student would fall below 12 credits. Of note is that the schedule of Fig. 15 is identical with that of z = 4, which suggests that in some cases with m = 3, z = 4 could suffice to constrain the shape to optimality.

Finally, with our Undeclared freshman, we observe a similar pattern to the Math junior. Though this student has more required courses than the Engineering senior, they are also afforded more options in terms of sections, and thus we were able to get an output that fits their preferred time quite well. We ran this profile with a preferred time period of 7am - 1pm. In Fig. 13, with z = 1, we see that the model schedules all required courses, along with a course outside of the Core. It arbitrarily schedules a Friday lab section. Also of note is that this schedule is the same as that of Fig. 10, which indicates that z = 1 has hardly any effect. In Fig. 16, with z = 5, we see that this non-Core course, occurring in the afternoon, is removed. To avoid the Friday course penalty, the lab is swapped to an equivalent Thursday section. This schedule appears to be at optimality for this student, as all courses . For this run, the schedule of Fig. 16 was not generated until z = 5. z = 2 through 4 resulted in the same schedules as z = 1.

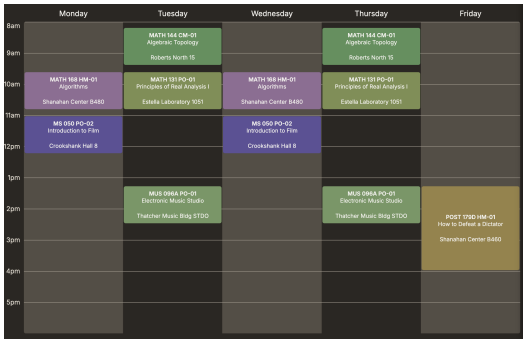


Figure 11: Profile 1 (Math Junior) Schedule generated by Shape Optimization Model 3.3.4 with $m = 3, z = 1, t = 0700, f = 1300$

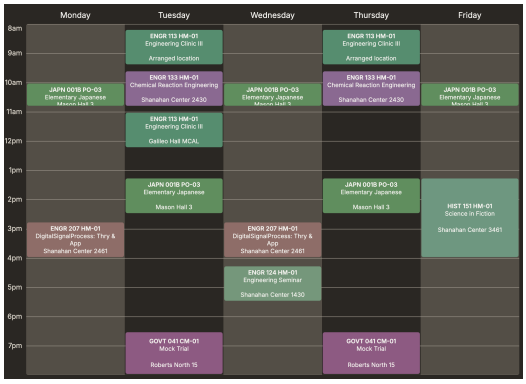


Figure 12: Profile 2 (Engineering Senior) Schedule generated by Shape Optimization Model 3.3.4 with $m = 3, z = 1, t = 1400, f = 2100$.

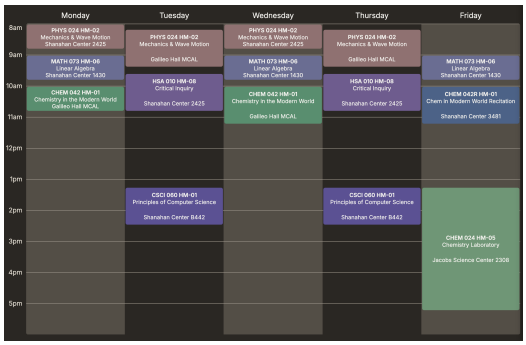


Figure 13: Profile 3 (Undecl. Freshman) Schedule generated by Shape Optimization Model 3.3.4 with $m = 3, z = 1, t = 0700, f = 1300$.

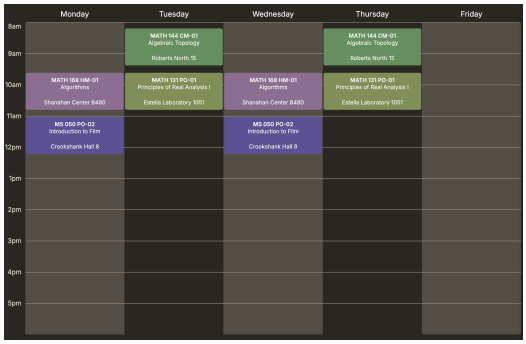


Figure 14: Profile 1 (Math Junior) Schedule generated by Shape Optimization Model 3.3.4 with $m = 3, z = 5, t = 0700, f = 1300$.

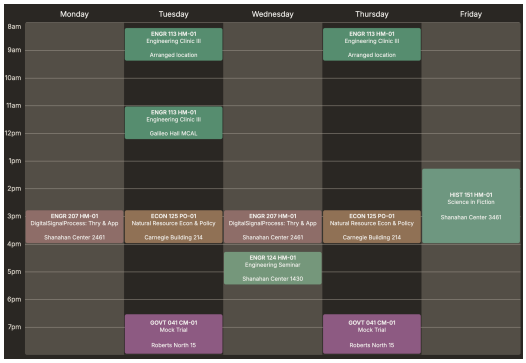


Figure 15: Profile 2 (Engineering Senior) Schedule generated by Shape Optimization Model 3.3.4 with $m = 3, z = 5, t = 1400, f = 2100$.

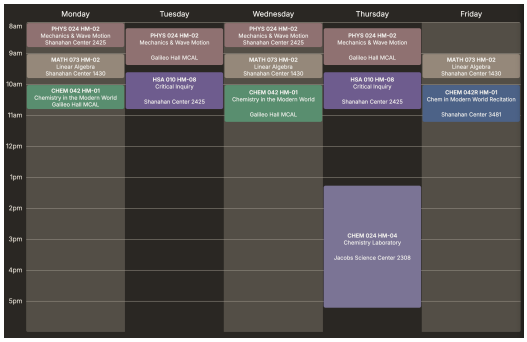


Figure 16: Profile 3 (Undecl. Freshman) Schedule generated by Shape Optimization Model 3.3.4 with $m = 3, z = 5, t = 0700, f = 1300$.

3.5 Conclusions and Recommendations

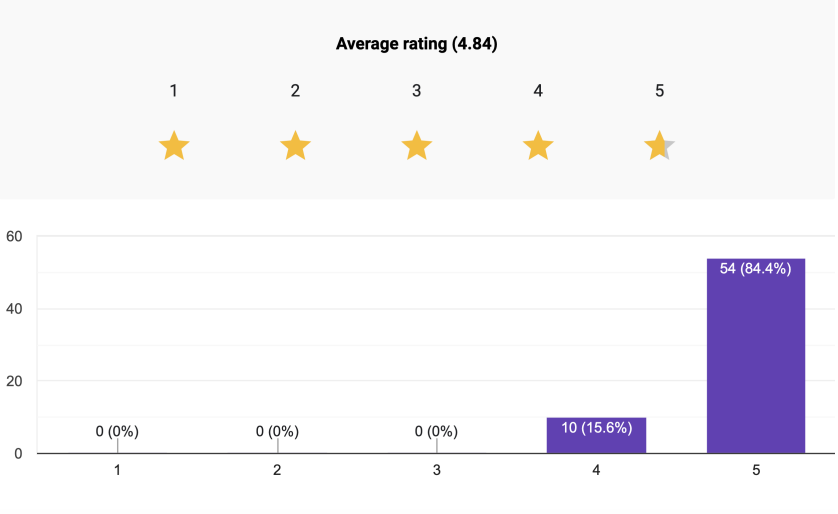
Course registration priorities can be modeled and adjusted by user- defined parameters in the *Student Interest Maximization* model. Based on the outputs modeled for a typical Harvey Mudd Math major, we found that setting the major multiplier to $m = 3$ resulted in a more balanced schedule between HSA courses and major requirements. A similar trend appeared in the Senior Engineering major profile as well. Therefore, we recommend this input parameter as a setting for students who seek to fulfill major requirements while pursuing personal interests. For distance and accessibility, we find that an adherence to both necessitates values of $\delta = 9$ and $y_c = 3$, though y_c should be adjusted according to the HSA rankings. If there is an HSA listed that is not accessible, y_c should be set to a value higher than that HSA ranking in order to be prioritized over that HSA. Depending on how strongly students consider course times in course registration, the penalty parameter in the schedule shape optimization model should be set accordingly. If students want as strong of an adherence to their shape preferences as possible, a value of z = 5 seems to suffice for most uses. z = 4 is a more relaxed option, which at times achieves the same schedule of z = 5, but is more flexible. To maintain a certain credit course load, we recommend that this penalty be set to $p_c = 3$, such that students will get a schedule that fits both interest and convenience. Amongst the various formulations, a moderate, halfway range for each weight parameter is best to accurately gauge interest without having significant tradeoffs with credit load and course areas.

4 Appendices

4.1 Student Survey

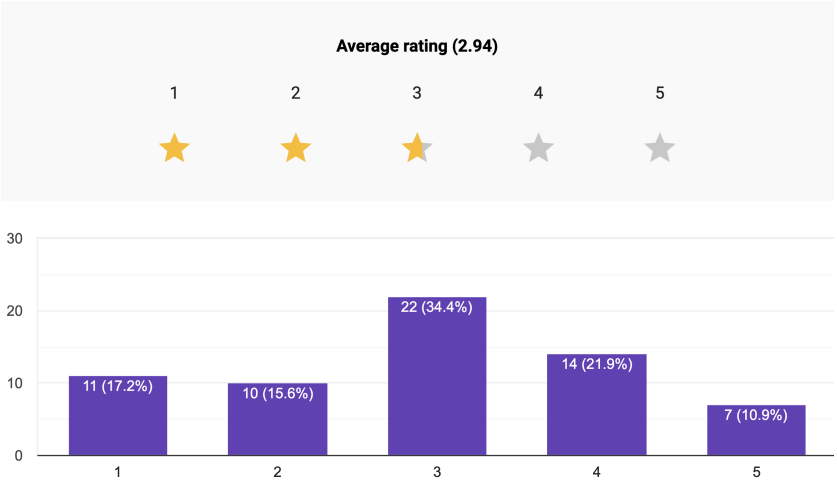
Our student-facing survey, sent to Harvey Mudd students of Linde Dorm, Case Dorm, Drinkward Dorm, and Atwood dorm, included the following required questions, with their aggregate results included:

1. How important is Hyperschedule to your class-picking process? (1 is least important, 5 is most important)



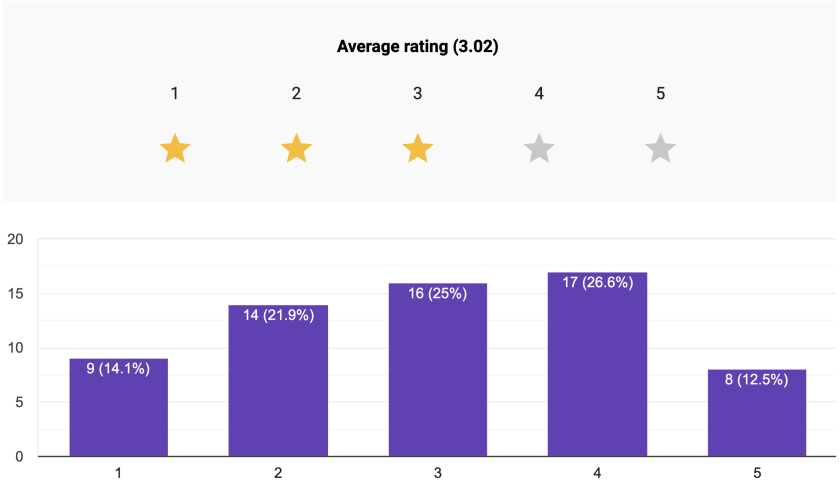
Variance: 0.1339285714

2. How important is avoiding morning classes for your class-picking process? (1 is least important, 5 is most important)



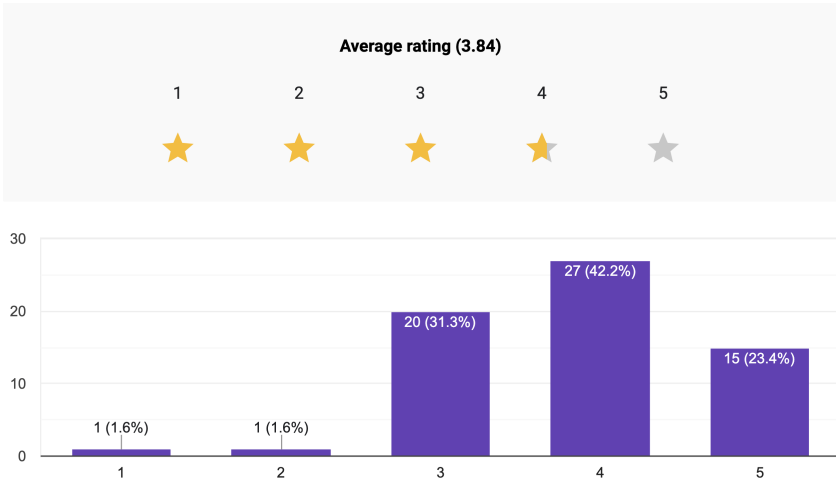
Variance: 1.51984127

3. How important is avoiding evening classes for your class-picking process? (1 is least important, 5 is most important)



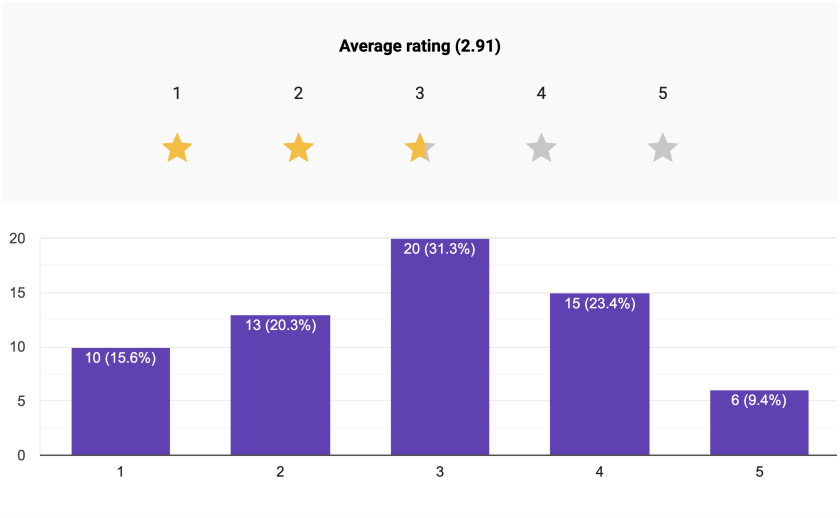
Variance: 1.571180556

4. How much does the professor teaching the section affect your class-picking process? (1 is least important, 5 is most important)



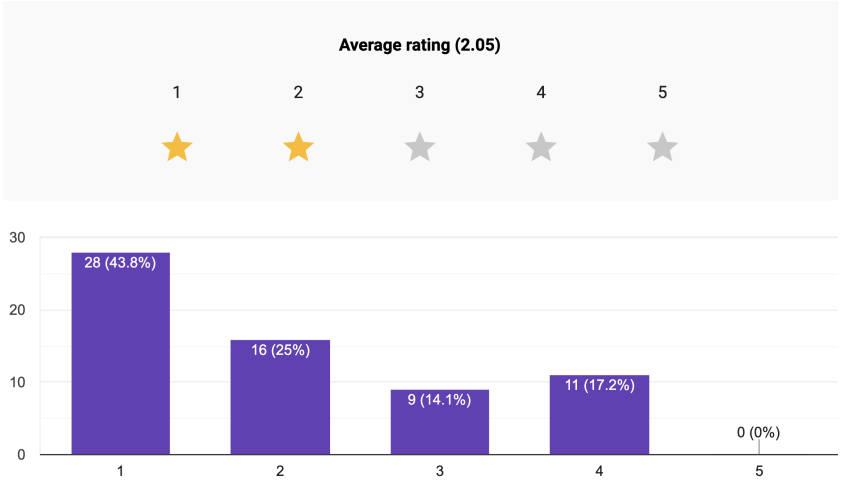
Variance: 0.7371031746

5. How important is RateMyProfessors.com to your class-picking process? (1 is least important, 5 is most important)



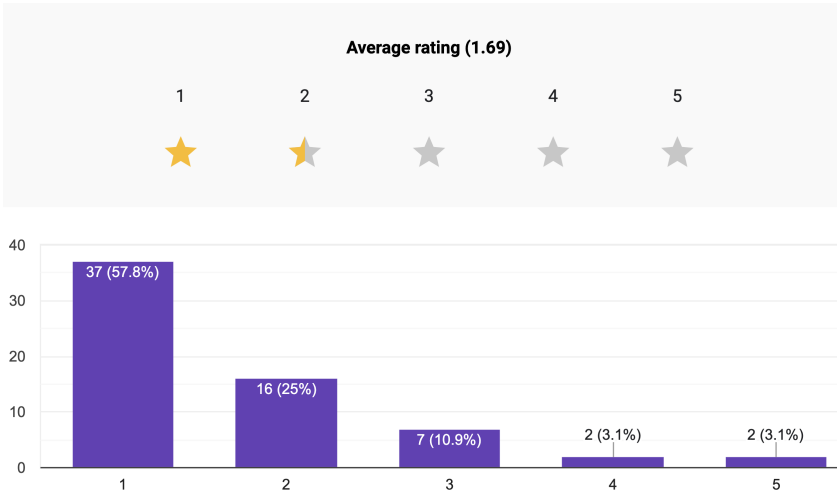
Variance: 1.451388889

6. How important is distance between classes to your class-picking process? (given that you have enough time to make it to class on-time) (1 is least important, 5 is most important)



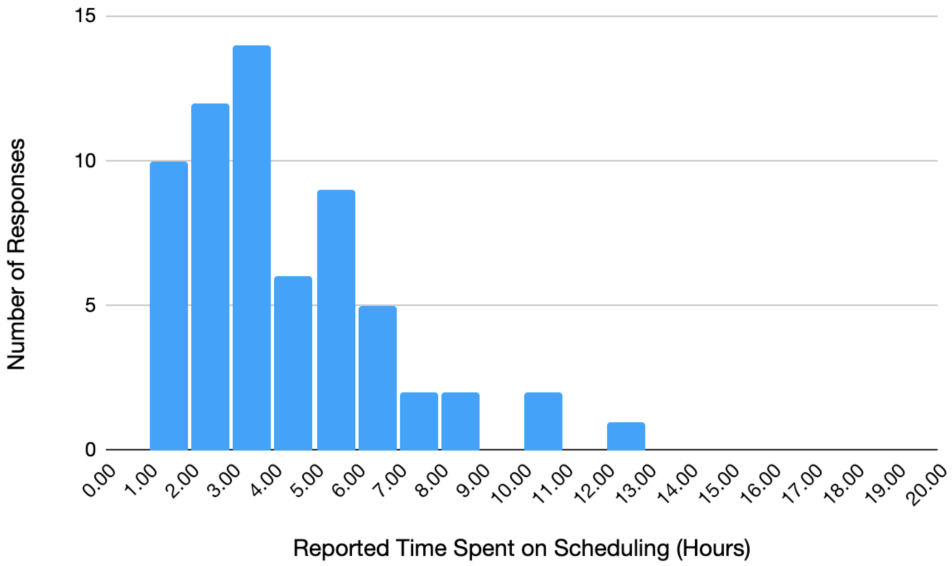
Variance: 1.283482143

7. How important is classroom accessibility to your class-picking process? (1 is least important, 5 is most important)



Variance: 1.011904762

8. How many hours would you estimate you spend on formulating a course schedule each semester?



Variance: 9.583085317

With the following optional, free-response questions:

9. What other factors are important to you when selecting classes?

We received 29 responses to this question. Some of the recurring factors in this set of responses were following major requirements, avoiding Friday courses, and whether or not the students' friends were taking the course.

10. What is the hardest part of coming up with your own schedule?

    We received 33 responses to this question. The overwhelmingly common response from students was that avoiding overlapping times was the most difficult part of creating a schedule.

4.2 Professor Survey

Our professor-facing survey, sent to a subset of professors from each academic discipline, contained the following questions, where respondents could choose which to engage with. We recieved four responses from professors.

1. What tools do you use to help students with course registration and what tools do you wish you had?
2. What pain points and/or flaws in the system do you / your advisees encounter during scheduling?
3. What seems to be the most important factors to your advisees when selecting courses?
4. What advice do you give to your advisees when they are designing their schedules?
5. What do you think about RateMyProfessor? Do you know your score or ever look at it?

4.3 Student Interest Maximization - Base Model

Sets:

- Courses ($\mathcal{C}$) := list of all offered courses (with section numbers)
- Course Codes := list of all course codes without regard to section numbers
- Days := set of days 'monday', 'tuesday', 'wednesday', 'thursday', 'friday'
- Time Periods := set of 5 minute periods between 00:00 and 23:00

Parameters:

- $r_c \in \{0, 1\}$:= Equals 1 if course c is **required** for the major.
- $m \geq 0$:= Scalar weight applied to **major**-required courses to balance with HSA ranking points
- $n \in \{0, 1\}$:= Equals 1 if the student **needs** HSA courses - HSA rankings should be weighted in equation; 0 otherwise.
- h_c := Student **HSA** ranking (interest) for course c (higher = more desirable).
- $v_c \geq 0$:= Number of credits awarded by course c.
- $\ell = 12$:= Minimum (**lower** bound) number of credits to be a full-time student

- u = 18 := Maximum (**upper** bound) number of credits to avoid overloading
- $i_c \in$ Course Codes := Course code **identifier** for course c
- $b_{c,t,d} \in \{0, 1\}$:= Equals 1 is course c is in session (**busy**) at time period t on day d

Decision Variables

- $x_c \in \{0, 1\}$:= 1 if course c is selected, 0 otherwise.

$$
\begin{array}{lll}
\textbf{maximize} & \displaystyle\sum_{c \in \mathcal{C}} x_c v_c \left(r_c\, m + n\, h_c\right) & \text{Objective Function} \\
\textbf{subject to} & \displaystyle\sum_{c \in \mathcal{C}} x_c\, v_c \geq \ell & \text{Lower bound on credits taken} \\
& \displaystyle\sum_{c \in \mathcal{C}} x_c\, v_c \leq u & \text{Upper bound on credits taken} \\
& \displaystyle\sum_{c \in \mathcal{C}:\ \mathrm{i}_c = j} x_c \leq 1 & \forall j \in \text{Course Codes (one section per course code)} \\
& \displaystyle\sum_{c \in \mathcal{C}} x_c\, b_{c,t,d} \leq 1 & \forall t \in \text{Time Periods}, d \in \text{Days (one course per time period)} \\
& x_c = 0 & \forall c \in \mathcal{C} \text{ where } v_c = 0 \text{ (do not include 0 credit courses)} \\
& x_c \in \{0, 1\} & \forall c \in \mathcal{C} \text{ (binary course selection)}
\end{array}
\tag{1}
$$

4.4 Distance Minimization and Accessibility Schedule

Model inherits all of sets, parameters, and decision variables from *Student Interest Maximization* model. It additionally has the following:

Sets:

- Buildings := list of all buildings courses are located

Parameters:

- $\mathrm{d}_{a,b} \geq 0$:= Distance between building a and building b in miles
- $a_k \in \{0, 1\}$:= Accessibility entrance indicator of building k
- $g_c \in$ Buildings := Building of course location
- $s_c, e_c \in \mathbb{Z}$:= Start/end times of course c
- $A \geq 0$:= Penalty for inaccessible buildings
- $\delta \geq 0$:= Penalty weight per mile for travel distance between courses

Decision Variables

- $y_c \geq 0$:= Accessibility penalty applied to c
- $q_c \geq 0$:= Travel time penalty applied to c
- $\rho_{i,j} \geq 0$:= Penalty for distance between courses i and j

maximize $\sum_{c \in \mathcal{C}} x_c v_c \left(r_c m + n h_c - y_c - q_c\right) \; - \; \delta \sum_{\substack{i,j \in \mathcal{C} \\ i \neq j}} \rho_{i,j}$

subject to

$\sum_{c \in \mathcal{C}} x_c v_c \geq \ell$	(Lower bound on credits)
$\sum_{c \in \mathcal{C}} x_c v_c \leq u$	(Upper bound on credits)
$\sum_{c \in \mathcal{C}: i_c = j} x_c \leq 1$	$\forall j \in$ CourseCodes (One section per course)
$\sum_{c \in \mathcal{C}} x_c b_{c,t,d} \leq 1$	$\forall t \in$ Time Periods, $d \in$ Days (No time conflicts)
$x_c = 0$	$\forall c \in \mathcal{C}$ when $v_c = 0$ (Exclude 0-credit courses)
$y_c = A$	$\forall c \in \mathcal{C}$ when $a_{g_c} = 0$ (penalty if no accessible entrance)
$y_c = 0$	$\forall c \in \mathcal{C}$ when $a_{g_c} = 1$ (Accessible building, no penalty)
$\rho_{i,j} = d_{g_i,g_j}\, x_i x_j$	$\forall\, i, j \in \mathcal{C}$ when $0 < s_j - e_i < 20$ (travel-window distance penalty)
$\rho_{i,j} = 0$	$\forall\, i, j \in \mathcal{C}$ when $s_j - e_i \leq 0 \ \vee \ s_j - e_i \geq 20$ (no travel penalty)
$x_c \in \{0, 1\}$	$\forall\, c \in \mathcal{C}$ (binary course selection)
$q_i, \rho_{i,j} \geq 0$	$\forall\, i, j \in \mathcal{C}$ (Non-negativity)

4.5 Schedule Shape Optimization

Model inherits all of sets, parameters, and decision variables from *Student Interest Maximization* model. It additionally has the following:

Parameters:

- $\mathrm{s}_c \in$ Time Periods := start time of course c
- $\mathrm{e}_c \in$ Time Periods := end time of course c
- $\mathrm{t} \in$ Time Periods := if a course starts before time t, it will be penalized
- $\mathrm{f} \in$ Time Periods := if a course ends after time f, it will be penalized
- $\mathrm{z} \geq 0$:= size of the penalty that is applied to a course outside of the time bounds

Variables:

- p_c := penalty applied course c

Below is the objective function and constraints for this model:

$$
\begin{aligned}
\textbf{maximize} \quad & \sum_{c \in \mathcal{C}} x_c v_c \left((\mathrm{r}_c\, m + n\, h_c) - p_c \right) && \text{Objective Function} \\
\textbf{subject to} \quad & \sum_{c \in \mathcal{C}} x_c\, v_c \geq \ell && \text{Lower bound on credits taken} \\
& \sum_{c \in \mathcal{C}} x_c\, v_c \leq u && \text{Upper bound on credits taken} \\
& \sum_{c \in \mathcal{C}:\ \mathrm{i}_c = j} x_c \leq 1 && \forall j \in \text{Course Codes (one section per course code)} \\
& \sum_{c \in \mathcal{C}} x_c\, b_{c,t,d} \leq 1 && \forall t \in \text{Time Periods}, d \in \text{Days (one course per time period)} \\
& x_c = 0 && \forall c \in \mathcal{C} \text{ where } v_c = 0 \text{ (do not include 0 credit courses)} \\
& p_c = z && \forall c \in \mathcal{C} \text{ where } s_c < t \text{ or } e_c > f \text{ (set penalty on)} \\
& p_c = 0 && \forall c \in \mathcal{C} \text{ where } s_c \geq t \text{ or } e_c \leq f \text{ (set penalty off)} \\
& x_c, p_c \geq 0 && \forall c \in \mathcal{C} \text{ (non-negativity)}
\end{aligned}
\tag{2}
$$

4.6 Simulated Student Profiles

We have devised three representative student profiles for our sensitivity analysis. The first of which applies to the fall semester, while the second and third apply to the spring semester. Importantly, our "Core requirements" are strictly enforced—the generated schedule must include them. However, "major requirements" and "HSA rankings" are incentivized, but not strictly enforced.

4.6.1 Profile 1: Mudd Junior Math major

This profile describes a junior Math major from Mudd, with the following requirements:
And the following HSA rankings:

Major Requirements
Real Analysis - MATH131 HM-01 FA2025 / MATH131 PO-01 FA2025
Abstract Algebra - MATH171 CM-01 FA2025 (any section)
Algorithms - MATH168 HM-01 FA2025
Algebraic Topology - MATH144 CM-01 FA2025
Combinatorial Mathematics - MATH103 PO-01 FA2025

HSA Rankings
(5) **Electronic Music Studio** - MUS 096A PO-01 FA2025
(4) **How to Defeat a Dictator** - POST179D HM-01 FA2025
(3) **Introduction to Film** - MS 050 PO-01 FA2025 / MS 050 PO-02 FA2025
(2) **Intro to Environmental Analysis** - EA 010 PO-01 FA2025 / EA 010 PZ-01 FA2025 / EA 010 SC-01 FA2025
(1) **Oppression and Liberation** - PSYC179U HM-01 FA2025

Notably, this student has no strictly required courses at this semester in their major, so no "Core"

classes are identified.

4.6.2 Profile 2: Mudd Senior Engineering major

This profile describes a senior Engineering major from Mudd, with the following requirements:

Core Requirements
Engineering Clinic III - ENGR113 HM-01 SP2025
Engineering Seminar - ENGR124 HM-01 SP2025

Major Requirements
Mechanical Design - ENGR177 HM-01 SP2025
Analog Circuit Design - ENGR151 HM-01 SP2025
Digital Signal Processes - ENGR207 HM-01 SP2025
Chemical Reaction Engineering - ENGR133 HM-01 SP2025
Human-Centered Design - ENGR180 HM-01 SP2025

HSA Rankings
(5) **Science in Fiction** - HIST151 HM-01 SP2025
(4) **Elementary Japanese** - JAPN001B PO-01 SP2025
(4) **Elementary Japanese** - JAPN001B PO-02 SP2025
(4) **Elementary Japanese** - JAPN001B PO-03 SP2025
(3) **Mock Trial** - GOVT041 CM-01 SP2025
(2) **Critical Geography of Silicon Valley** - GEOG179H HM-01 SP2025
(1) **Natural Resource Econ and Policy** - ECON125 PO-01 SP2025

In this case, we set a list of upper-division Engineering electives as the major requirements, as elective credits are all the student has left to complete within their major beyond Clinic and Seminar, which must be on their schedule. Additionally, this student still needs to complete their HSA breadth requirement, so various HSA courses in different disciplines are ranked. For variety, we imagine this student has some experience in Japanese language from High School, and can place into the spring continuation of Elementary Japanese.

4.6.3 Profile 3: Mudd Second-semester Freshman

This profile describes an undeclared freshman from Mudd, with the following requirements:

Core Requirements
Chemistry in the Modern World - CHEM042 HM SP2025 (any section)
Chemistry in the Modern World Recitation - CHEM042R HM SP2025 (any section)
Chemistry Laboratory - CHEM024 HM SP2025 (any section)
Linear Algebra - MATH073 HM SP2025 (any section)
Mechanics and Wave Motion - PHYS024 HM SP2025 (any section)

Major Requirements
Intro Engineering Design/Manufacturing - ENGR004 HM-01 SP2025 (any section)
Intro Engineering Design/Manufacturing Lab - ENGR004L HM-01 SP2025 (any section)
Principles of Computer Science - CSCI060 HM-01 SP2025 (any section)
Intro to Computational Biology - MCBI118B HM-01 SP2025

HSA Rankings
(5) **Critical Inquiry Section 8** - HSA 010 HM-08 SP2025
(4) **Critical Inquiry Section 15** - HSA 010 HM-15 SP2025
(3) **Critical Inquiry Section 4** - HSA 010 HM-04 SP2025
(2) **Critical Inquiry Section 13** - HSA 010 HM-13 SP2025
(1) **Critical Inquiry Section 1** - HSA 010 HM-01 SP2025

Because the freshman schedule is largely predetermined, the majority of this student's courses are placed in the strict "Core requirements" set. Notably, we apply a slight abuse of notation within the "major requirements" set. Because this student is not yet sure what their major will be, the courses listed under this heading are not current requirements, but rather, a selection of courses that may contribute to their major in the future. For this reason, they are not strictly required. We include them as examples of courses that second-semester freshmen often consider adding to their schedule. Similarly, though HSA010 is a required course for this student, it is listed under "HSA rankings", which are not strictly required, either. However, as we show in our sensitivity analysis, we are able to ensure that one of the sections is chosen reliably with our selection of the majorMultiplier m.

5 Bibliography

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